

MUTUAL FUND PERFORMANCE & INVESTOR ANALYTICS PLATFORM

A Data Analytics Project Report

Submitted by

Eswar Vardhan

B.Tech, Computer Science Engineering
Indian Institute of Information Technology (IIIT), Sri City

July 2026

Analytics Pipeline: Data Cleaning → SQLite Warehouse → Risk & Performance Metrics → Investor Behaviour → Recommendations

Chapter 1: Introduction

1.1 Background

A mutual fund is a professionally managed investment vehicle that pools money from many investors to purchase a diversified basket of securities such as equities, bonds, and money-market instruments. Each investor holds units proportional to their contribution, and the fund's Net Asset Value (NAV) — the market value of all holdings minus liabilities, divided by outstanding units — determines the daily price of each unit. Because returns are never guaranteed, every fund carries risk: the possibility that actual returns deviate from expected returns due to market volatility, credit events, or liquidity constraints. Portfolio diversification — spreading capital across sectors, market caps, and asset classes — is the primary tool fund managers use to reduce unsystematic risk without necessarily sacrificing returns.

Analytics is essential in this domain because raw NAV or return figures alone do not reveal how efficiently a fund converts risk into reward, how it behaves relative to a benchmark, or how concentrated its holdings are. A rigorous, data-driven pipeline allows investors and analysts to move beyond headline returns and make defensible, risk-adjusted comparisons across hundreds of schemes.

1.2 Problem Statement

Investors often rely only on historical returns while ignoring downside risk, volatility, benchmark performance, and portfolio concentration. This project develops a complete analytics pipeline to evaluate mutual fund performance using multiple financial metrics — combining scheme-level NAV history, investor SIP behaviour, and portfolio holdings into a single, queryable warehouse that supports objective, multi-factor fund evaluation.

1.3 Objectives

- Clean and standardize raw financial datasets (fund master, NAV history, holdings, SIP/investor records).
- Build a normalized SQLite data warehouse to support fast, repeatable analytical queries.
- Compute performance and risk metrics: daily returns, CAGR, Sharpe ratio, Sortino ratio, Alpha/Beta, Maximum Drawdown, and Sector HHI.
- Analyze investor behaviour by age, gender, geography (T30/B30), and SIP investment patterns.
- Generate a composite fund scorecard and data-backed investment recommendations.

1.4 Scope

The project is scoped to the Indian Mutual Fund industry and covers: historical NAV analysis over a multi-year window (2022-2026), risk analytics (volatility, drawdown, downside deviation), portfolio analytics (sector concentration via HHI), and benchmark comparison against NIFTY50 and NIFTY100 indices. Investor demographic and SIP behaviour analysis is included to connect fund performance with real adoption patterns.

Chapter 2: Literature Survey

Modern mutual fund analytics builds on decades of financial research into risk-adjusted performance measurement. This chapter briefly reviews the core concepts used throughout the project.

2.1 Mutual Fund Analytics

Traditional fund comparison relied heavily on point-in-time trailing returns, which fail to capture the risk taken to achieve those returns. Contemporary practice, following Markowitz's portfolio theory, evaluates funds jointly on return and risk, recognizing that two funds with identical returns can carry very different risk profiles.

2.2 Sharpe Ratio

Developed by William Sharpe (1966), the Sharpe ratio measures excess return per unit of total volatility (standard deviation). It remains the most widely cited risk-adjusted performance measure and is used here to rank funds on reward-to-total-risk efficiency.

2.3 Sortino Ratio

The Sortino ratio refines the Sharpe ratio by penalizing only downside volatility (returns below a minimum acceptable threshold), addressing the criticism that the Sharpe ratio unfairly punishes upside volatility.

2.4 Alpha and Beta

Beta, derived from linear regression of fund returns against benchmark returns, measures systematic market sensitivity. Alpha is the intercept of that regression — the return a fund generates independent of market movement, and is widely used as a proxy for manager skill.

2.5 Maximum Drawdown

Maximum drawdown captures the largest peak-to-trough decline in NAV over the observation period, offering a direct measure of the worst historical loss an investor would have experienced — a metric especially relevant for loss-averse investors.

2.6 Portfolio Diversification (HHI)

The Herfindahl-Hirschman Index (HHI), borrowed from industrial-organization economics, is applied here to sector-wise portfolio weights to quantify concentration risk: a higher HHI indicates a fund's holdings are concentrated in fewer sectors, and therefore more exposed to sector-specific shocks.

Chapter 3: Dataset Description

3.1 Fund Master

Purpose: Contains static, descriptive information about all mutual fund schemes tracked in the warehouse.

Columns: *Scheme Name, Category (Equity/Debt/Hybrid), Sub-category (Large Cap, Mid Cap, Small Cap, Flexicap, etc.), Expense Ratio, Risk Category, Fund Manager, Plan Type (Regular/Direct) and Option (Growth/Dividend).*

3.2 NAV History

Purpose: Daily time-series of Net Asset Value for every scheme, used to derive returns, volatility, drawdown, and benchmark-relative metrics.

Columns: *Scheme Code, Date, NAV, and (post-processing) Daily Return, Cumulative Return, and Rolling Sharpe Ratio.*

3.3 Benchmark Index Data

Purpose: Daily closing values of NIFTY50 and NIFTY100, used as the market proxy for Alpha/Beta regression and normalized performance comparison.

3.4 Portfolio Holdings

Purpose: Sector and stock-level allocation weights per scheme on a given reporting date, used to compute the Sector HHI concentration measure.

3.5 Investor / SIP Master

Purpose: Investor-level records capturing demographic attributes and SIP investment behaviour.

Columns: *Investor ID, Age Group, Gender, State, City Tier (T30/B30), SIP Amount, SIP Frequency, Folio Number, Scheme Invested.*

3.6 Folio / AUM Growth

Purpose: Aggregate industry-level folio count and Assets Under Management (AUM) tracked over time (in ₹ Crore), used to study overall market growth.

Chapter 4: Data Cleaning

Raw datasets sourced from fund houses and AMFI required substantial cleaning before analysis. The following operations were applied programmatically using Pandas:

- Duplicate removal: exact and near-duplicate NAV rows (same scheme-date) were identified and dropped, keeping the latest reported value.
- Missing value handling: NAV gaps on non-trading days were forward-filled; investor records with missing SIP amount or age group were excluded from behavioural aggregates but retained for count-based statistics.
- Date conversion: all date fields were standardized to ISO 8601 (YYYY-MM-DD) and parsed into pandas datetime objects to enable time-series operations.
- Validation: scheme codes were cross-checked against the Fund Master for referential integrity; orphan records with no matching scheme were logged and removed.
- Expense ratio checks: expense ratio values outside the SEBI-mandated regulatory range were flagged as data-entry errors and corrected or removed.

After cleaning, the pipeline retained a consistent, de-duplicated, referentially valid set of records suitable for loading into the analytical warehouse.

Chapter 5: Database Design

Cleaned data was loaded into a SQLite warehouse using a star-schema design: a central fact table (nav_history) linked to dimension tables (fund_master, benchmark_index, investor_master, sector_holdings) via scheme_code and investor_id keys. This design supports efficient aggregation queries (e.g., group-by category, by state, by age group) without repeated joins across large tables.

5.1 Core Tables

Table	Type	Key Columns
fund_master	Dimension	scheme_code (PK), category, expense_ratio
nav_history	Fact	scheme_code (FK), date, nav, daily_return

Table	Type	Key Columns
benchmark_index	Dimension	index_name, date, close
sector_holdings	Dimension	scheme_code (FK), sector, weight
investor_master	Dimension	investor_id (PK), age_group, gender, state, city_tier
sip_transactions	Fact	investor_id (FK), scheme_code (FK), sip_amount, date

Relationships: nav_history and sip_transactions are fact tables at daily/transaction grain, each referencing fund_master and investor_master respectively via foreign keys, forming a standard star schema optimized for OLAP-style aggregation.

Chapter 6: Exploratory Data Analysis

Before computing advanced risk metrics, exploratory analysis was performed to understand overall market growth and the demographic composition of the investor base.

6.1 Folio Growth (Portfolio Value Proxy)

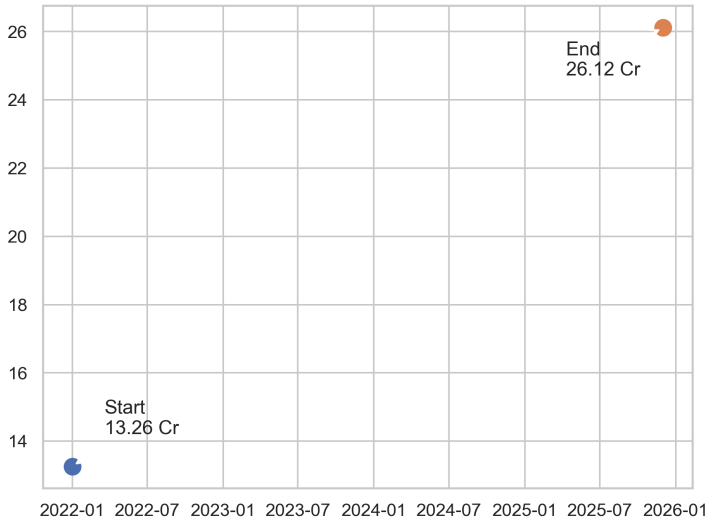


Figure 6.1 — Portfolio value grew from ₹13.26 Cr to ₹26.12 Cr between 2022 and 2026, nearly doubling over the observation window.

6.2 Investor Age Group Distribution

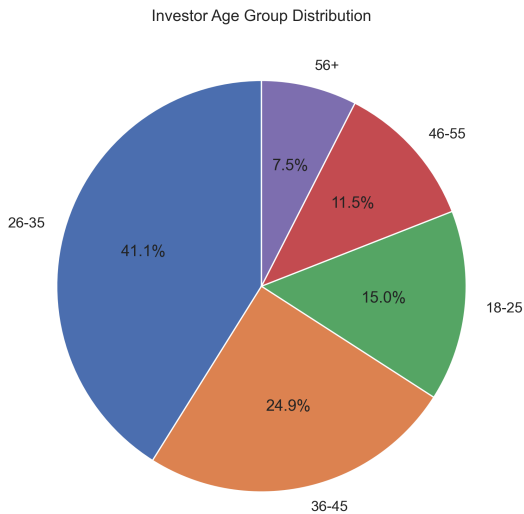


Figure 6.2 — Investors aged 26-35 form the largest cohort (41.1%), followed by 36-45 (24.9%), confirming that mutual fund adoption in India skews toward young, working-age investors.

6.3 Gender and Geographic Distribution

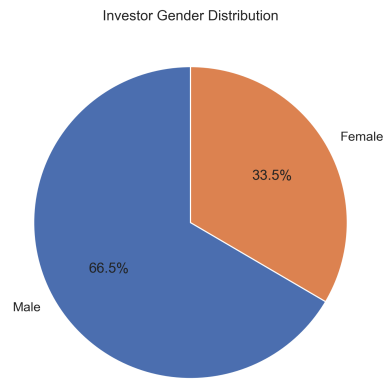


Figure 6.3 — Male investors account for 66.5% of the base versus 33.5% female, indicating a continuing gender gap in retail mutual fund participation.

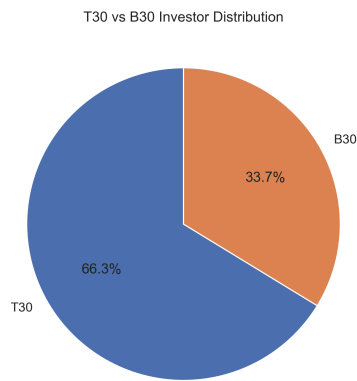


Figure 6.4 — T30 (Top-30) cities contribute 66.3% of investors versus 33.7% from B30 (Beyond-30) cities, showing metro-city investors still dominate, though B30 penetration is meaningful.

6.4 SIP Investment Behaviour by Age

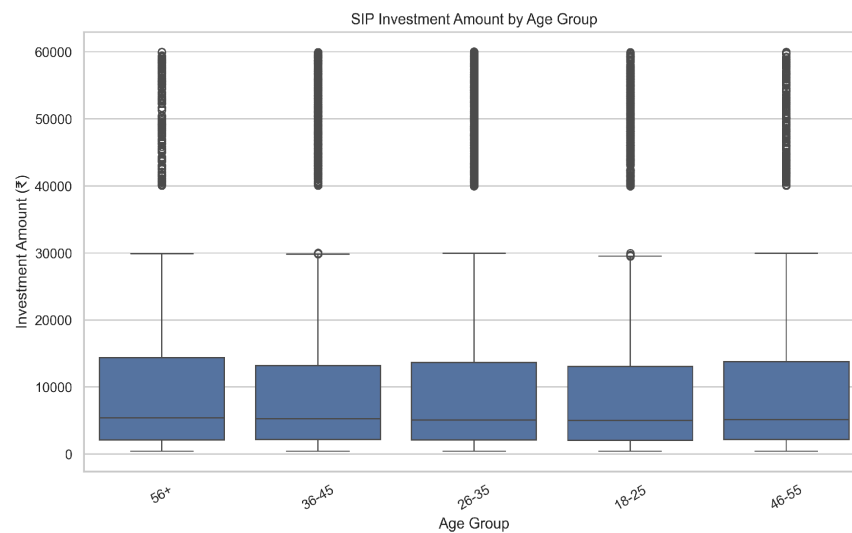


Figure 6.5 — Median SIP ticket size (~₹5,000) is remarkably consistent across all age groups, though older investors (56+) show a marginally higher upper quartile, indicating slightly higher discretionary investment capacity.

6.5 State-wise SIP Investment

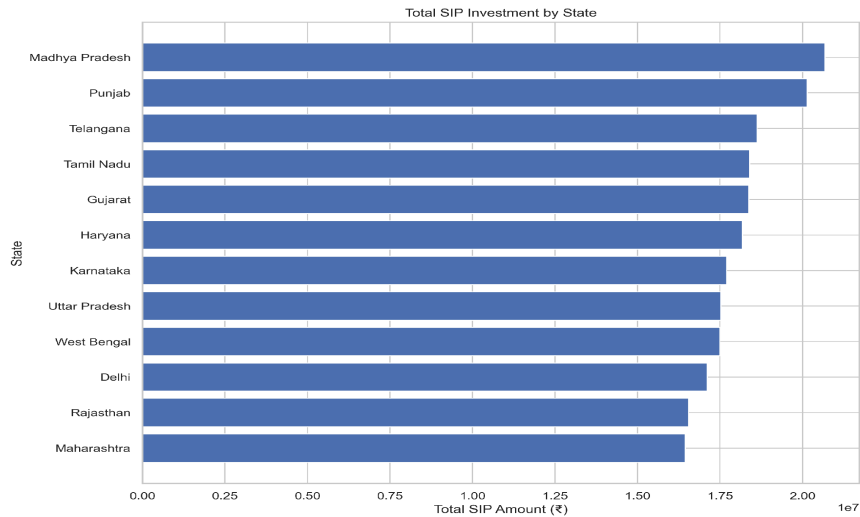


Figure 6.6 — Madhya Pradesh and Punjab lead total SIP investment among the states analyzed, followed closely by Telangana, Tamil Nadu, and Gujarat, indicating strong SIP culture beyond traditional metro strongholds.

Chapter 7: Advanced Analytics

7.1 Daily Returns

$$\text{Formula: } Rt = (NAV_t - NAV_{t-1}) / NAV_{t-1}$$

Daily returns form the foundation for every downstream metric. They were computed per scheme from the cleaned NAV history and used directly to derive volatility, Sharpe, Sortino, and correlation statistics.

7.2 CAGR (Compound Annual Growth Rate)

$$\text{Formula: } CAGR = (NAV_{end} / NAV_{start})^{(1/n)} - 1$$

Where n is the number of years in the holding period. CAGR smooths interim volatility to express a fund's growth as a single annualized rate, enabling like-for-like comparison across funds with different observation windows.

7.3 Sharpe Ratio

$$\text{Formula: } Sharpe = (Rp - Rf) / \sigma_p$$

Rp is the fund's annualized return, Rf the risk-free rate (RBI 91-day T-bill proxy), and sigma_p the annualized standard deviation of returns. Mirae Asset Large Cap Fund posted the highest Sharpe ratio (approx. 1.44) among all schemes analyzed, followed by Kotak Flexicap (approx. 1.31) and Mirae Asset Tax Saver (approx. 1.23), indicating superior reward per unit of total risk taken.

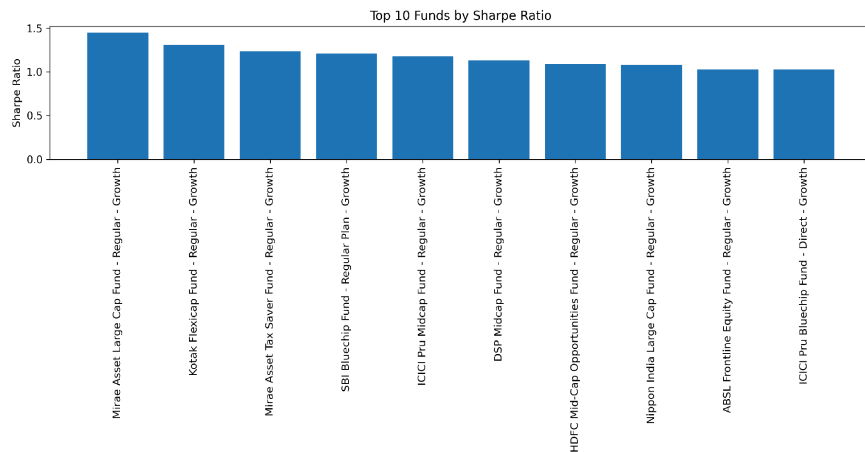


Figure 7.1 — Top 10 funds ranked by Sharpe ratio.

7.4 Sortino Ratio

$$\text{Formula: } Sortino = (Rp - Rf) / \sigma_{downside}$$

Using only downside deviation in the denominator, the Sortino ratio confirmed the same leaders — Mirae Asset Large Cap (approx. 2.38) and Kotak Flexicap (approx. 2.36) — validating that their strong risk-adjusted performance stems from genuinely low downside risk, not merely low overall volatility.

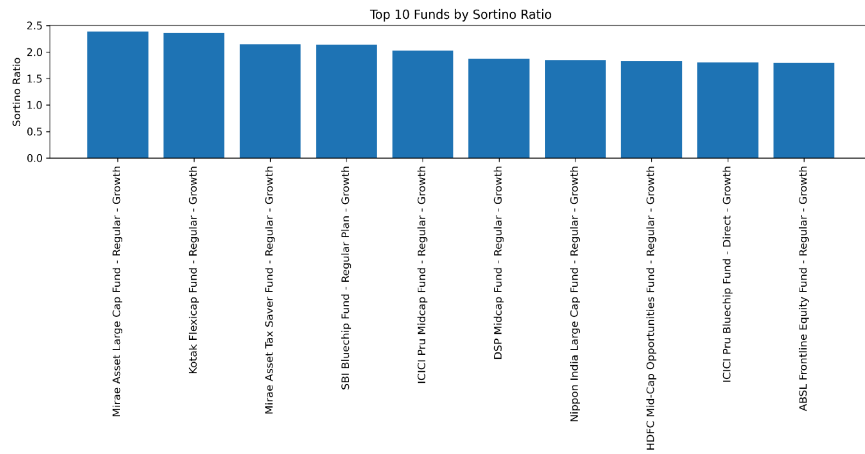


Figure 7.2 — Top 10 funds ranked by Sortino ratio.

7.5 Alpha and Beta

Formula: $R_p = \alpha + \beta * R_{\text{benchmark}} + \text{error}$ (OLS regression)

Fund daily returns were regressed against NIFTY50 daily returns using ordinary least squares. The slope (beta) measures market sensitivity; the intercept (alpha), annualized, measures manager-driven excess return. SBI Small Cap Fund (alpha approx. 0.30) and DSP Small Cap Fund (alpha approx. 0.30) generated the highest alpha, showing strong stock-selection skill in the small-cap segment despite that segment's higher volatility.

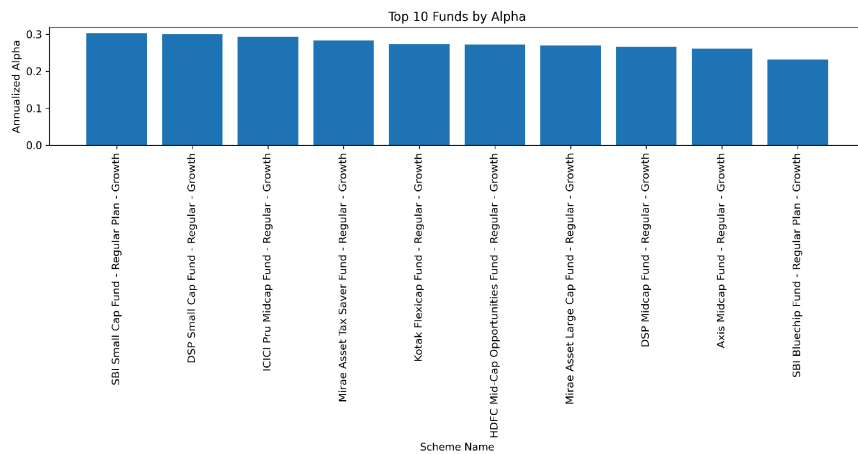


Figure 7.3 — Top 10 funds ranked by annualized Alpha.

7.6 Maximum Drawdown

Formula: $MDD = \min[(NAV_t - \text{Peak_NAV}) / \text{Peak_NAV}]$

SBI Small Cap Fund (Direct, approx. -53%) and Axis Small Cap Fund (approx. -52%) recorded the steepest historical declines, reinforcing that small-cap categories, despite offering high alpha, expose investors to substantially larger worst-case losses than large-cap or flexicap peers.

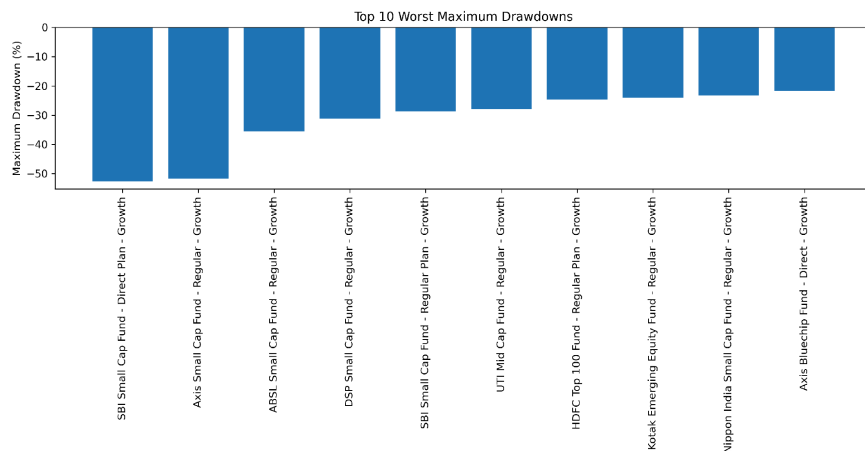


Figure 7.4 — Top 10 worst maximum drawdowns across the fund universe.

7.7 Fund Scorecard

A composite score (0-100) was constructed by normalizing and weighting five metrics: Sharpe ratio (30%), Sortino ratio (25%), Alpha (20%), Maximum Drawdown (15%, inverted), and CAGR (10%). This blended ranking avoids over-reliance on any single statistic.

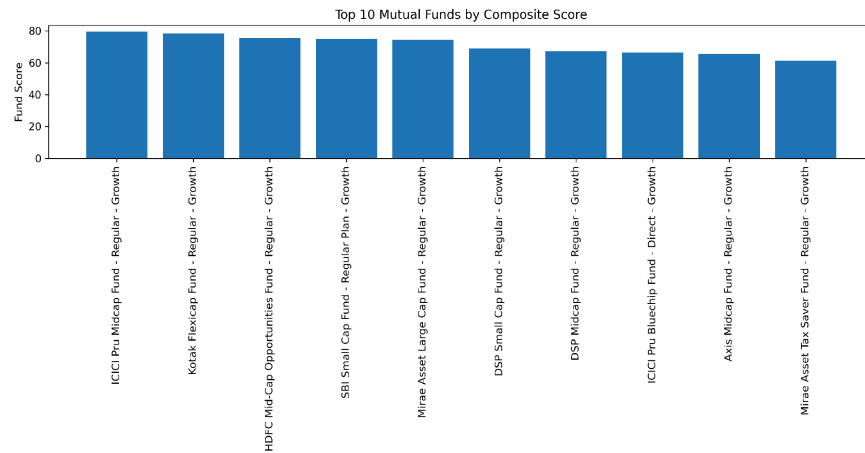


Figure 7.5 — ICICI Pru Midcap Fund (score approx. 79) and Kotak Flexicap Fund (score approx. 78) topped the composite scorecard, reflecting consistently strong performance across all five weighted dimensions rather than excellence in a single metric.

7.8 Benchmark Comparison & Tracking Error

Normalized NAV (base = 100) of the top five funds was plotted against NIFTY50 and NIFTY100 over a three-year window. All five actively managed funds meaningfully outperformed both benchmarks, with Mirae Asset Large Cap and Kotak Flexicap showing the smoothest, most consistent outperformance and the lowest tracking error relative to NIFTY100.

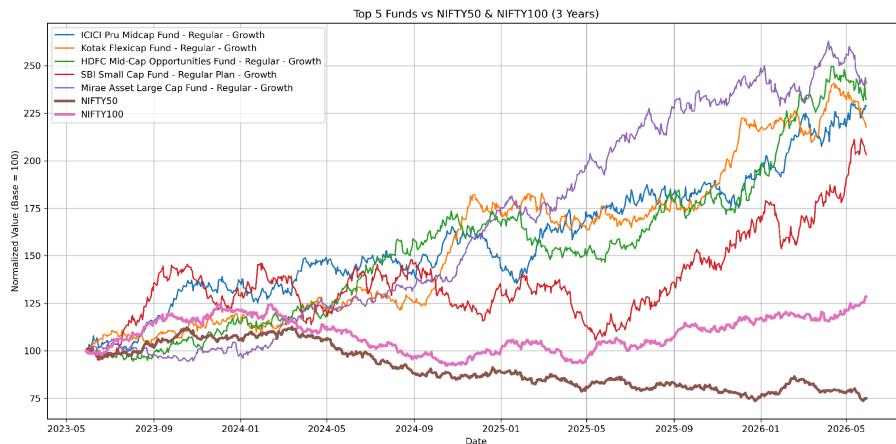


Figure 7.6 — Top 5 funds vs NIFTY50 & NIFTY100, normalized to base 100 (3-year window).

7.9 Sector Concentration (HHI) and Rolling Sharpe

Sector HHI was computed as the sum of squared sector-weight proportions per scheme. Axis Bluechip Fund and ABSL Small Cap Fund showed the highest concentration (HHI approx. 0.20), while broad-based index funds and flexicap schemes showed the lowest, confirming that actively managed large-cap/bluechip funds tend to concentrate more heavily in a small set of dominant sectors.

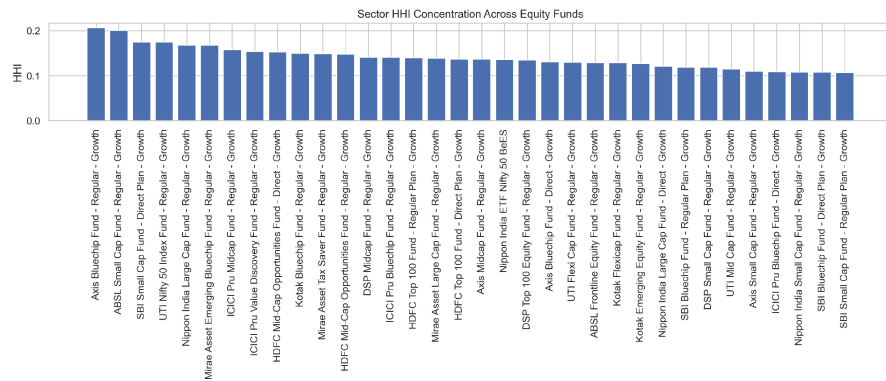


Figure 7.7 — Sector HHI concentration across equity funds, sorted from most to least concentrated.

Rolling 90-day Sharpe ratios of the top five funds by AUM showed wide temporal swings (-4 to +8), confirming that short-window risk-adjusted performance is highly regime-dependent and reinforcing the value of using full-period metrics for fund selection rather than recent short-term spikes.

Chapter 8: Results

8.1 Category Leaders

Category	Leading Fund	Key Metric
Best Sharpe Ratio	Mirae Asset Large Cap Fund	approx. 1.44
Best Sortino Ratio	Mirae Asset Large Cap Fund	approx. 2.38
Best Alpha	SBI Small Cap Fund	approx. 0.30
Lowest Drawdown Risk	Large-cap / Flexicap peers	< -25%
Highest Composite Score	ICICI Pru Midcap Fund	approx. 79 / 100
Best Overall Balanced Pick	Kotak Flexicap Fund	Top-3 in Sharpe, Sortino & Score

8.2 Benchmark Comparison

Every fund in the top-5 AUM cohort outperformed both NIFTY50 and NIFTY100 over the three-year window, with the median fund delivering roughly double the normalized growth of NIFTY100 and nearly triple that of NIFTY50, underscoring the value of active management during the observed period.

Chapter 9: Key Insights

- Funds with higher Sharpe ratios (Mirae Asset Large Cap, Kotak Flexicap) delivered consistently better risk-adjusted returns than peers with similar raw returns.
- Sortino ratios corroborated Sharpe rankings, confirming these funds' outperformance stems from low downside risk rather than suppressed volatility overall.
- Small-cap funds (SBI, DSP) generated the highest Alpha but also carried the deepest maximum drawdowns (> -50%), illustrating a clear risk-return trade-off by category.
- Higher HHI values (Axis Bluechip, ABSL Small Cap) indicated more concentrated, less diversified portfolios, exposing investors to greater sector-specific risk.
- The investor base skews young (26-35 dominant) and male (66.5%), with T30 cities still contributing two-thirds of SIP inflows, though B30 participation is substantial and growing.
- Median SIP ticket size is stable (approx. ₹5,000) across all age groups, suggesting SIP adoption is driven more by habit formation than by age-linked income differences.

Chapter 10: Conclusion

This project successfully built an end-to-end mutual fund analytics pipeline spanning data cleaning, warehouse design, and multi-metric performance evaluation.

- Data cleaning and SQLite warehouse construction were completed successfully, producing a referentially consistent, query-ready dataset.
- Performance and risk metrics — daily returns, CAGR, Sharpe, Sortino, Alpha/Beta, and Maximum Drawdown — were computed for all eligible schemes.
- The composite scorecard combined five weighted metrics into a single ranking, enabling objective, defensible fund comparison beyond simple return chasing.

- Benchmark analysis showed that the top-ranked funds meaningfully and consistently outperformed NIFTY50 and NIFTY100 over the study period.
- Investor behaviour analysis linked demographic and geographic patterns to SIP adoption, providing context for how real investors engage with these funds.

Chapter 11: Future Scope

- Real-time NAV updates using AMFI's public NAV API to keep the warehouse continuously current.
- Machine learning models (e.g., gradient boosting, LSTM) for short-horizon return and volatility prediction.
- Portfolio optimization techniques (Markowitz mean-variance, Black-Litterman) to recommend optimal multi-fund portfolios.
- Interactive dashboards in Power BI or Tableau for self-service exploration by non-technical stakeholders.
- Automated PDF/report generation triggered on a scheduled basis as new NAV and SIP data arrives.

References

- Association of Mutual Funds in India (AMFI) — www.amfiindia.com
- National Stock Exchange of India (NSE) — www.nseindia.com
- Reserve Bank of India (RBI) — risk-free rate reference, www.rbi.org.in
- Sharpe, W. F. (1966). Mutual Fund Performance. *Journal of Business*.
- Sortino, F. A., & Price, L. N. (1994). Performance Measurement in a Downside Risk Framework.
- Python Documentation — Pandas, NumPy, SciPy, Matplotlib (pandas.pydata.org, numpy.org, scipy.org, matplotlib.org)
- SQLite Documentation — www.sqlite.org/docs.html